

GridBalance V2 — Experiment Registry Log

Complete Record of All Candidate Optimization Experiments & Sealed-Test Metrics

experiment_id	dataset	hypothesis	feature_set	model	sampling_strategy	hyperparameters	cv_metric	validation_metric	test_metric	training_time	artifact	status	notes
EXP_SGCC_V2_01	SGCC	Re-evaluate frozen 18-feature XGBoost baseline benchmark	18 Features	XGBoost Classifier	Class-Weighted (10.72)	max_depth=7, n_estimators=200, lr=0.2	0.4029 F1	0.4037 F1	0.3592 F1 / 0.3114 PR-AUC	0.080s	models/sgcc_tuned/xgboost_reduced_best.joblib	BASELINE	Frozen primary baseline model
EXP_SGCC_V2_02	SGCC	LightGBM GBDT histogram binning on 18 features	18 Features	LightGBM Classifier	Class-Weighted (10.72)	default LGBM, scale_pos_weight=10.72	nan	0.3084 F1	0.3084 F1 / 0.2831 PR-AUC	0.501s	models/optimization_v2/lgb_sgcc_v2_02.joblib	EVALUATED	Default LightGBM baseline
EXP_SGCC_V2_03	SGCC	25-trial Optuna CV tuning for LightGBM	18 Features	Optuna LightGBM	Class-Weighted (10.72)	{'n_estimators': 300, 'max_depth': 10, 'num_leaves': 115, 'learning_rate': 0.020448998598217673, 'subsample': 0.6573565746297758, 'colsample_bytree': 0.7126598162330608, 'reg_alpha': 0.006257706947781621, 'reg_lambda': 0.001007934337527647}	0.3545 F1	0.3602 F1	0.3245 F1 / 0.2938 PR-AUC	245.490s	models/optimization_v2/lgb_sgcc_v2_03.joblib	EVALUATED	Optuna 5-fold CV optimization
EXP_SGCC_V2_04	SGCC	Threshold optimization on Validation predictions (grid [0.1, 0.9])	18 Features	Optuna LightGBM (Tuned Thresh)	Threshold=0.58	Optimal Threshold=0.58	0.3545 F1	0.3694 F1	0.3366 F1 / 0.2938 PR-AUC	0.010s	models/optimization_v2/lgb_sgcc_v2_03.joblib	EVALUATED	Optimal validation threshold = 0.58
EXP_SGCC_V2_05	SGCC	Add 2 domain interaction features (volatility_ratio, missing_density)	20 Features	LightGBM (Interactions)	Class-Weighted (10.72)	20 features + Optuna params	nan	0.3452 F1	0.3247 F1 / 0.2901 PR-AUC	4.681s	models/optimization_v2/lgb_sgcc_v2_05.joblib	EVALUATED	Added domain interaction ratios
EXP_UCI_V2_01	UCI	Re-evaluate frozen 29-feature XGBoost regressor baseline	29 Features	XGBoost Regressor	Chronological Split (75/12.5/12.5)	n_estimators=300, max_depth=6, lr=0.05	10,365.77 kWh MAE	3987.48 kWh MAE	4503.51 kWh MAE / 1.98% SMAPE	0.027s	models/uci_forecasting/xgb_forecasting_frozen.joblib	BASELINE	Frozen primary baseline forecasting model
EXP_UCI_V2_02	UCI	LightGBM Regressor histogram split performance on 29 features	29 Features	LightGBM Regressor	Chronological Split	n_estimators=300, lr=0.05	nan	3926.32 kWh MAE	4391.08 kWh MAE / 1.91% SMAPE	1.620s	models/optimization_v2/lgb_uci_v2_02.joblib	EVALUATED	Default LightGBM regressor
EXP_UCI_V2_03	UCI	25-trial Optuna hyperparameter optimization for LightGBM Regressor	29 Features	Optuna LightGBM Regressor	Chronological Split	{'n_estimators': 394, 'max_depth': 7, 'num_leaves': 110, 'learning_rate': 0.05057777827888316, 'subsample': 0.8189752954165729, 'colsample_bytree': 0.9610077195040908}	nan	3888.65 kWh MAE	4377.14 kWh MAE / 1.90% SMAPE	47.864s	models/optimization_v2/lgb_uci_v2_03.joblib	EVALUATED	Optuna tuned LightGBM Regressor

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EXP_UCI_V2_04	UCI	Add 4 seasonal Fourier & EMA features (ema_6, ema_24, sin/cos_month)	33 Features	LightGBM (33 Features)	Chronological Split	33 features + Optuna params	nan	3885.47 kWh MAE	4348.93 kWh MAE / 1.89% SMAPE	3.365s	models/optimization_v2/lgb_uci_v2_04.joblib	EVALUATED	Added EMA & Fourier monthly features